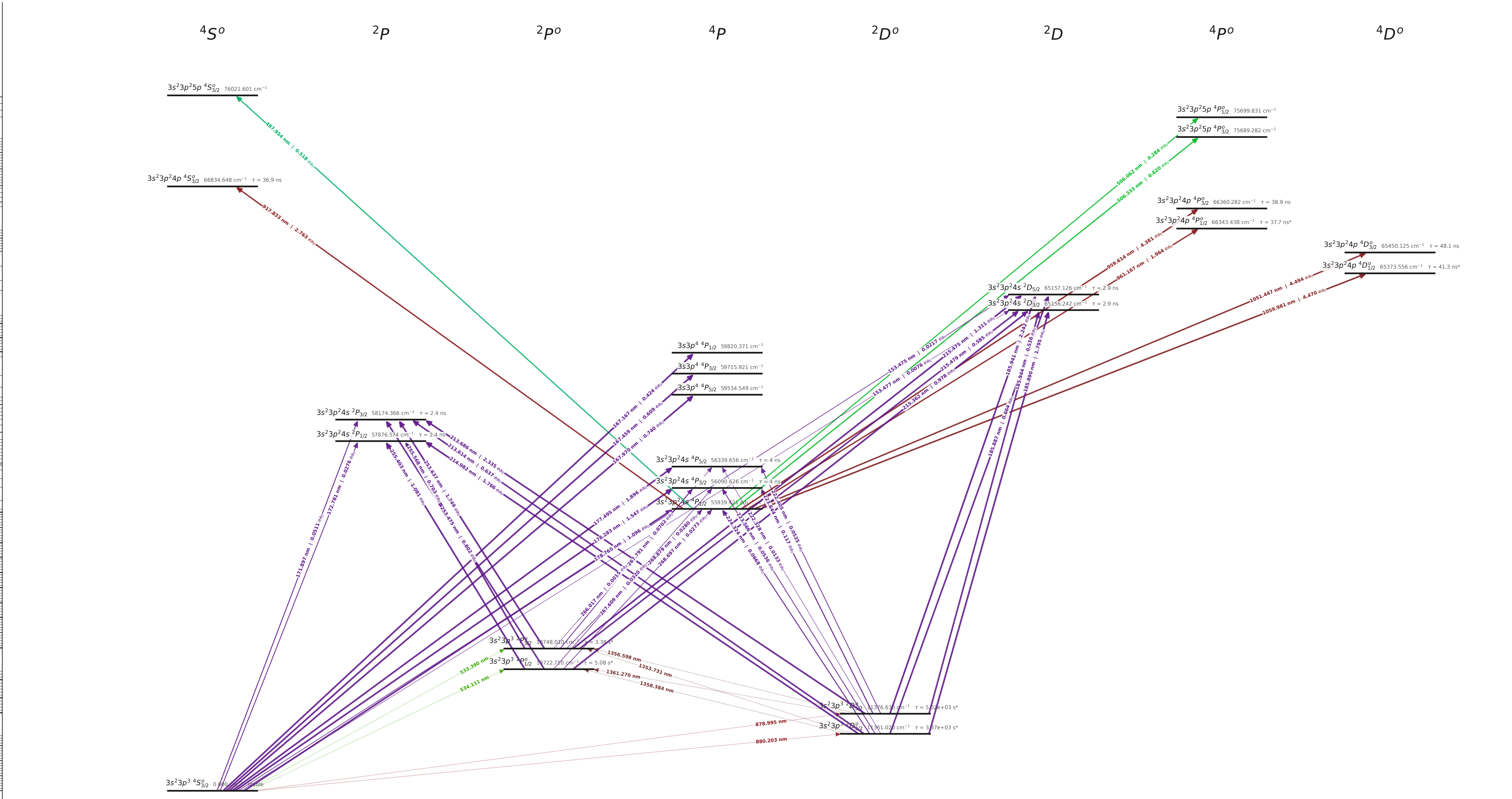


# Phosphorus

P (P I). Strongest classified transitions below 2 μm (NIST): 23 levels, 50 lines.

P<sup>+</sup> ionisation limit 84580.83 cm<sup>-1</sup> = 10.48669 eV (118.230 nm)



## Key transitions

| Transition                                                                                                            | λ vac (nm) | λ air (nm) | ν (THz)   | U  er  J'   (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|-----------------------------------------------------------------------------------------------------------------------|------------|------------|-----------|--------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| 3s <sup>2</sup> 3p <sup>3</sup> 4S <sub>3/2</sub> <sup>o</sup> - 3s3p <sup>4</sup> 4P <sub>1/2</sub>                  | 167.1671   | -          | 1793.3696 | 0.424                          | 3.900e+07            | -              | -          | 167.2 / 167.5 / 168.0 nm multiplet 3p3 4S°3/2 - 3s3p4 4P     |
| 3s <sup>2</sup> 3p <sup>3</sup> 4S <sub>3/2</sub> <sup>o</sup> - 3s3p <sup>4</sup> 4P <sub>3/2</sub>                  | 167.4595   | -          | 1790.2383 | 0.609                          | 4.000e+07            | -              | -          | 167.2 / 167.5 / 168.0 nm multiplet 3p3 4S°3/2 - 3s3p4 4P     |
| 3s <sup>2</sup> 3p <sup>3</sup> 4S <sub>3/2</sub> <sup>o</sup> - 3s3p <sup>4</sup> 4P <sub>5/2</sub>                  | 167.9697   | -          | 1784.8009 | 0.740                          | 3.900e+07            | -              | -          | 167.2 / 167.5 / 168.0 nm multiplet 3p3 4S°3/2 - 3s3p4 4P     |
| 3s <sup>2</sup> 3p <sup>3</sup> 4S <sub>3/2</sub> <sup>o</sup> - 3s <sup>2</sup> 3p <sup>2</sup> 4s 4P <sub>3/2</sub> | 177.4949   | -          | 1689.0204 | 1.896                          | 2.170e+08            | 39.79          | 86.8       | 177.5 / 178.3 / 178.8 nm resonance triplet 3p3 4S°3/2 - 4... |
| 3s <sup>2</sup> 3p <sup>3</sup> 4S <sub>3/2</sub> <sup>o</sup> - 3s <sup>2</sup> 3p <sup>2</sup> 4s 4P <sub>3/2</sub> | 178.2829   | -          | 1681.5547 | 1.547                          | 2.140e+08            | 39.79          | 85.6       | 177.5 / 178.3 / 178.8 nm resonance triplet 3p3 4S°3/2 - 4... |
| 3s <sup>2</sup> 3p <sup>3</sup> 4S <sub>3/2</sub> <sup>o</sup> - 3s <sup>2</sup> 3p <sup>2</sup> 4s 4P <sub>1/2</sub> | 178.7648   | -          | 1677.0216 | 1.096                          | 2.130e+08            | 39.79          | 85.2       | 177.5 / 178.3 / 178.8 nm resonance triplet 3p3 4S°3/2 - 4... |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>31</sup>P hyperfine structure (I = 1/2)

| Level                                                          | A (MHz)      | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|----------------------------------------------------------------|--------------|---------|--------------------------------------------------------|
| 3s <sup>2</sup> 3p <sup>3</sup> 4S <sub>3/2</sub> <sup>o</sup> | 55.055691(8) | -       | F=1: -68.82 F=2: +41.29                                |

Measured A, B constants; sources in the data file.